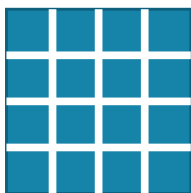
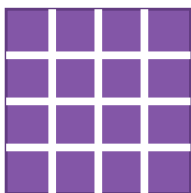


01 WEIGHT UPDATE

Frozen weights
 $W_0 \in \mathbb{R}^{(d \times k)}$



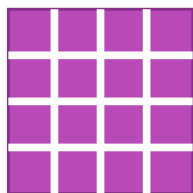
LoRA update
 $\Delta W = (\alpha/r)BA$



+

=

Adapted weights
 $W^* = W_0 + \Delta W$



The backbone stays frozen; training learns only the compact correction ΔW .

02 LOW-RANK DECOMPOSITION

B
 $d \times r$



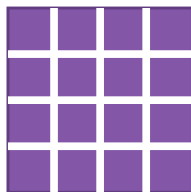
A
 $r \times k$



×

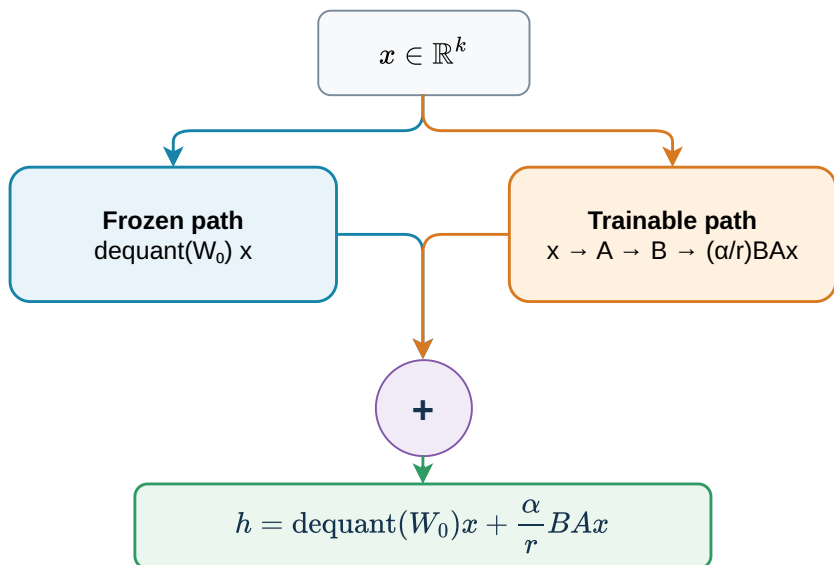
=

BA
 $d \times k$



$$r \ll \min(d, k) \implies \text{rank}(BA) \leq r$$

03 FORWARD PASS



Visual dimensions are schematic; the equations state the exact tensor shapes.